

CLOUD WEB SERVER SECURITY MIGRATION REPORT

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1. Executive Summary

Based on Step 1: Migration Objectives.

Global-Tech Solutions successfully migrated its legacy on-premises web server to a secure AWS Cloud environment (EC2). The primary objective was to enhance data availability and implement modern security controls, including automated backups, encrypted storage, and restricted network access, to protect sensitive customer information from unauthorized access and data loss.

2. Current vs. Target Environment Analysis

Based on Step 2: Environment Comparison.

Feature	On-Premises (Legacy)	AWS Cloud (Target)
Compute Instance	Physical Dell Server (Ubuntu 18.04)	AWS EC2 t3.medium (Ubuntu 22.04 LTS)
Network Security	Hardware Firewall (Static Rules)	AWS Security Groups (Stateful Inspection)
Data Encryption	None (Cleartext on Disk)	AWS EBS Encryption (AES-256)
Backup Strategy	Manual Weekly Tapes	AWS Backup (Daily Snapshots)

3. Security Implementation Steps

Based on Step 3: Deployment & Hardening.

- Infrastructure as Code:** Utilized Terraform to provision the EC2 instance, ensuring a repeatable and documented deployment process.
- Network Hardening:** Configured an AWS Security Group to allow only **Inbound TCP 80/443** from the public internet and **TCP 22** from the corporate VPN range. All other ports are restricted by default.
- IAM Role Assignment:** Applied the principle of least privilege by attaching an IAM role to the EC2 instance that allows only necessary access to S3 for log storage.
- SSH Security:** Disabled password-based authentication, enforcing the use of RSA Key Pairs for all administrative access.

4. Post-Migration Validation & Compliance

Based on Step 4: Verification.

The following tests were performed to ensure the new environment meets security standards:

- **Vulnerability Scan:** An Amazon Inspector scan was performed, resulting in zero "High" or "Critical" findings.
- **Connectivity Test:** Verified that the web server is reachable via HTTPS while direct SSH access is blocked from non-corporate IP addresses.
- **Backup Restoration:** Successfully performed a test restore from an EBS snapshot to verify data integrity and Recovery Time Objectives (RTO).